Effects of Light Wavelength on Plant Growth - Science Project Report

Document Type: School Project Report

Project Overview

Our science project explores the effects of different light wavelengths on plant growth. We hypothesized that blue light would promote faster growth compared to red and green light. This project was conducted over a period of 6 weeks in our school laboratory.

Materials and Methods

We used three identical plant pots with the same soil type and seed variety. Each pot was placed under different colored LED lights: blue, red, and green. We measured plant height daily and recorded observations about leaf color and overall health.

Experimental Setup

The experiment was conducted in a controlled environment with consistent temperature and humidity. Plants were watered equally and received 12 hours of light exposure daily. Measurements were taken at the same time each day to ensure consistency.

Results

After 6 weeks, plants under blue light showed 25% more growth compared to red light and 40% more than green light. Leaf color was also more vibrant under blue light. Plants under green light showed the slowest growth rate.

Analysis

Our results support the hypothesis that blue light promotes better plant growth. This is likely due to blue light's role in photosynthesis and chlorophyll production. The findings align with previous research on light spectrum effects on plant development.

Conclusion

Blue light significantly enhances plant growth compared to other wavelengths. This knowledge can be applied to indoor gardening and agricultural practices. Future experiments could explore different plant species and light intensity variations.